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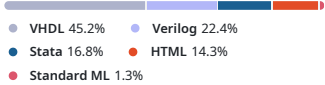
Releases

No releases published

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No packages published

Languages



		6 Commits
db		
incremental_db		
output_files		
simulation		
LICENSE		
README.md		
SRFLIPFLOPUSINGCASE.qpf		
SRFLIPFLOPUSINGCASE.qsf		
SRFLIPFLOPUSINGCASE.qws		
SRFLIPFLOPUSINGCASE.v		
SRFLIPFLOPUSINGCASE.v.bak		
Waveform.vwf		

📖 README

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SR-FLIPFLOP-USING-CASE

AIM:

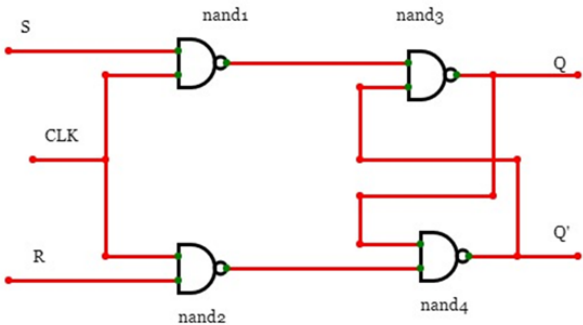
To implement SR flipflop using verilog and validating their functionality using their functional tables

SOFTWARE REQUIRED:

Quartus prime

THEORY

SR Flip-Flop SR flip-flop operates with only positive clock transitions or negative clock transitions. Whereas, SR latch operates with enable signal. The circuit diagram of SR flip-flop is shown in the following figure.



This circuit has two inputs S & R and two outputs Qtt & Qtt'. The operation of SR flipflop is similar to SR Latch. But, this flip-flop affects the outputs only when positive transition of the clock signal is applied instead of

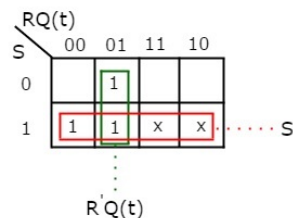
active enable. The following table shows the state table of SR flip-flop.

S	R	Q_{t+1}
0	0	Q_t
0	1	0
1	0	1
1	1	-

Here, Q_t & Q_{t+1} are present state & next state respectively. So, SR flip-flop can be used for one of these three functions such as Hold, Reset & Set based on the input conditions, when positive transition of clock signal is applied. The following table shows the characteristic table of SR flip-flop. Present Inputs Present State Next State

Present Inputs		Present State	Next State
S	R	Q_t	Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

By using three variable K-Map, we can get the simplified expression for next state, Q_{t+1} . The three variable K-Map for next state, Q_{t+1} is shown in the following figure.



The maximum possible groupings of adjacent ones are already shown in the figure. Therefore, the simplified expression for next state Q_{t+1} is $Q(t+1) = S + R'Q(t)Q(t+1) = S + R'Q(t)$

Procedure

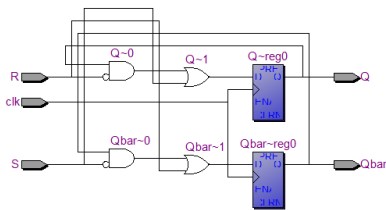
/* write all the steps involved */

PROGRAM . Developed by: Visveswarran Harikrishnan RegisterNumber: 212224110063

```
module exp6(S,R,clk,Q,Qbar);
input S,R,clk;
output reg Q;
output reg Qbar;
initial Q=0;
initil Qbar=1;
always @(posedge clk)
begin
Q=S \((`R)&Q);
end
endmodule
```



RTL LOGIC FOR FLIPFLOPS



TIMING DIGRAMS FOR FLIP FLOPS

